



Name: Cohort/Batch: Date: Score:

FAULT TOLERANCE PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Distributed Systems

TOTAL: 10 QUESTIONS

Q1 What is Fault Tolerance and what problem in Distributed Systems does it solve?

Latency Model

Q6 How does Fault Tolerance handle reliability and high availability?

Observability

Q2 What are the core components or concepts in Fault Tolerance?

Tolerance

Q7 What observability does Fault Tolerance provide out of the box?

Recycle

Q3 How does Fault Tolerance compare to its main alternatives?

Replication

Q8 What are common performance bottlenecks in Fault Tolerance?

Tolerance

Q4 How do you get started with Fault Tolerance for a new project?

Project

Q9 What security best practices apply when running Fault Tolerance?

Testing

Q5 What configuration options most affect Fault Tolerance's behavior in production?

SSO / IdP

Q10 What mistakes do teams commonly make when adopting Fault Tolerance?

Fault



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Fault Tolerance is a tool or platform

Mitigation

Fault Tolerance is a tool or platform that automates or simplifies a specific concern in the Distributed Systems domain, reducing manual effort and operational complexity for engineering teams.

A6 Fault Tolerance provides HA through leader election, observability

Fault Tolerance provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 Fault Tolerance has key building blocks that

Primitives

Fault Tolerance has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Fault Tolerance exposes metrics (Prometheus endpoint, structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

Fault Tolerance exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 Fault Tolerance trades off simplicity against feature

Hardening

Fault Tolerance trades off simplicity against feature richness differently than its alternatives. Choose Fault Tolerance when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install Fault Tolerance via its package manager

Gotchas

Install Fault Tolerance via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Fault Tolerance with the principle of

Assessment

Run Fault Tolerance with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency

Configuration

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.